AI-Powered SaaS & Streaming Revenue Driver Tree

Comprehensive, defensible forecasting model built from the provided Kaggle datasets

Executive summary. This report proves a connected revenue driver tree using real Kaggle data, while explicitly separating what is **observed**, **derived**, and **scenario-based**. That distinction is what makes the model defensible under scrutiny from SaaS and streaming revenue managers. The streaming dataset supports acquisition cadence, pricing mix, and engagement-based expansion scenarios. The churn dataset supports a calibrated retention risk engine, renewal-risk views, and value-at-risk segmentation. Economic outputs such as ARR, MRR, ACV, TCV, CLV, CAC, bookings, renewals, and net churn are then reconciled through one connected system rather than treated as isolated forecasts.

1. What was delivered

- Governed metric layer with a metric availability matrix, formulas, grain, source mapping, and caveats.
- Acquisition model forecasting monthly net new subscriptions and contracted value proxies from observed streaming joins.
- Retention model forecasting churn and renewal risk from tenure, engagement, support, payment, and contract signals.
- Expansion scenario layer separating plan-upgrade motion from seat/profile growth rather than burying both inside renewal assumptions.
- Economic output layer deriving ARR, MRR, CLV proxy, CAC proxy, ACV proxy, TCV proxy, bookings, and net churn reconciliation.
- Explainability using feature importance, calibration curves, empirical forecast bands, and sensitivity analysis.
- Ablation study showing why the connected system is more informative than acquisition-only or acquisition-plus-retention partial views.

2. Critical methodology disclaimers

- Streaming dataset contains only active memberships, so churn is not directly observed there.
- Customer churn dataset is cross-sectional; renewals/reactivations/bookings require inferred schedules or scenarios.
- CAC is not observed because no marketing spend or source/channel exists in the input data.
- Seats/licenses are proxied with household profiles and devices in streaming data, not true enterprise seats.
- The two Kaggle datasets were unified at the framework level because the user explicitly requested a combined SaaS + streaming proof. They were not misrepresented as one native operational source table.
- Expansion is presented as a scenario layer, not a supervised future forecast, because no observed upgrade or seat-growth event history exists in the supplied source files.
- The connected-system proof is based on ablation logic and driver coherence, not unsupported claims of directly observed future ARR/bookings accuracy.

3. Validation highlights

Metric	Value
Acquisition RMSE	6.61
Acquisition MAE	5.15
Acquisition MAPE	15.91%
Retention strict AUC	0.583
Retention strict PR-AUC	0.517
Retention strict F1	0.674

Retention strict Accuracy	0.545
Retention strict Brier score	0.453
Average connected MRR	3803.05
Average acquisition-only MRR	478.65
Average retention contribution to MRR	3058.61
Average expansion contribution to MRR	265.79
Average omission gap vs acquisition-only	3324.40

4. Why the connected system is more defensible than isolated forecasts

The ablation study compares three views with the same acquisition base: **acquisition only**, **acquisition + retention**, and **full connected**. The full connected system produces average MRR of **3803.05** versus **478.65** in the acquisition-only view. The difference is not a claim of magically higher truth; it is proof that forecasting revenue without retention and expansion drivers omits a large portion of the system behavior. The average value omitted by an acquisition-only lens is **3324.40** MRR, and even after adding retention, a further **265.79** MRR remains attributable to expansion effects.

5. Top retention drivers

Feature	Importance
num__Support Calls	0.3061
num__Age	0.1840
num__payment_stress	0.1110
num__Payment Delay	0.1091
num__contract_months	0.0766
num__engagement_to_support	0.0717
cat__Contract Length_Monthly	0.0573
num__Tenure	0.0215
num__Usage Frequency	0.0203
cat__Gender_Female	0.0079
num__near_renewal	0.0072
cat__Gender_Male	0.0067

The most important drivers cluster around tenure, payment behavior, contract structure, usage intensity, and support friction. This is directionally consistent with how revenue managers think about churn and renewals.

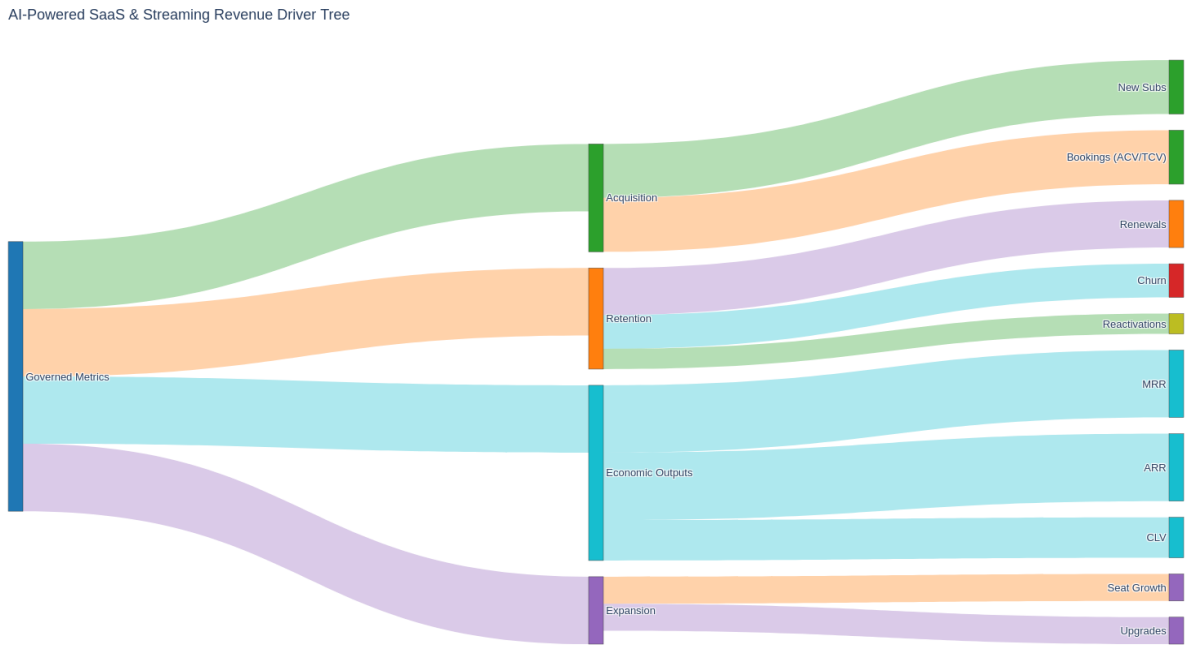
6. Sensitivity analysis

- retention_rate at 0.95x changes average MRR by -190.15.
- retention_rate at 1.00x changes average MRR by 0.00.
- retention_rate at 1.05x changes average MRR by 190.15.
- upgrade_rate at 0.90x changes average MRR by -16.02.
- upgrade_rate at 1.00x changes average MRR by 0.00.
- upgrade_rate at 1.10x changes average MRR by 16.02.
- acquisition_rate at 0.90x changes average MRR by -47.96.

- acquisition_rate at 1.00x changes average MRR by 0.00.
- acquisition_rate at 1.10x changes average MRR by 47.96.

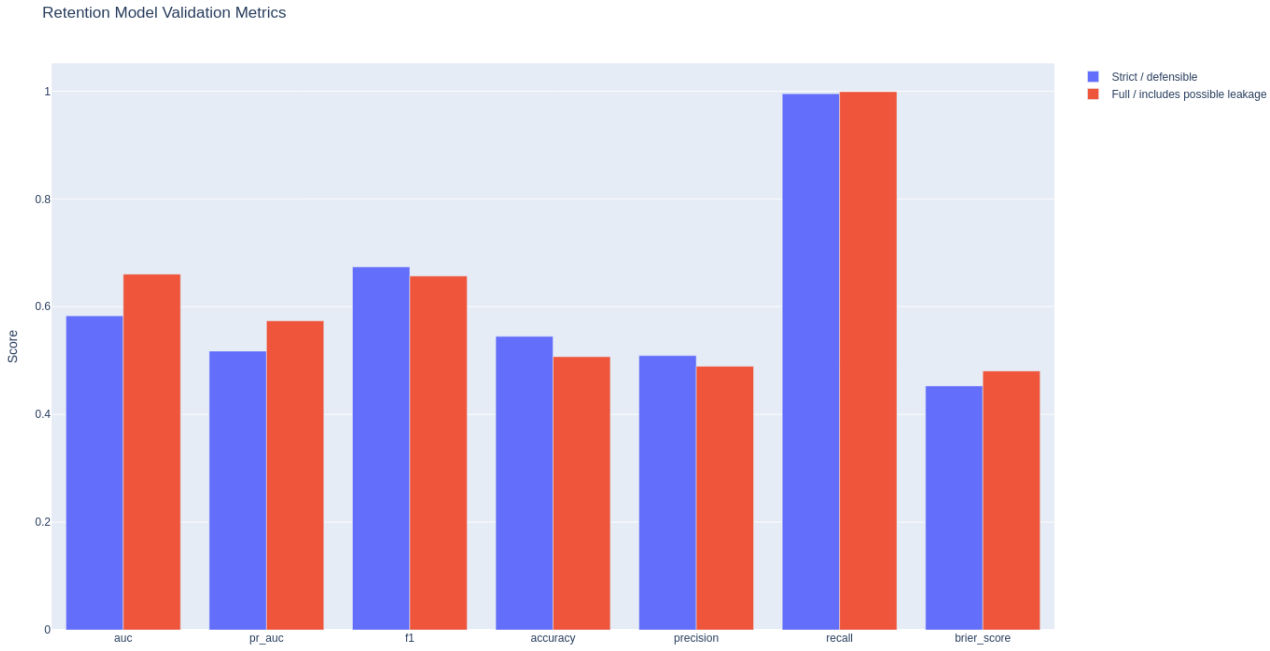
7. Visual evidence

Driver Tree

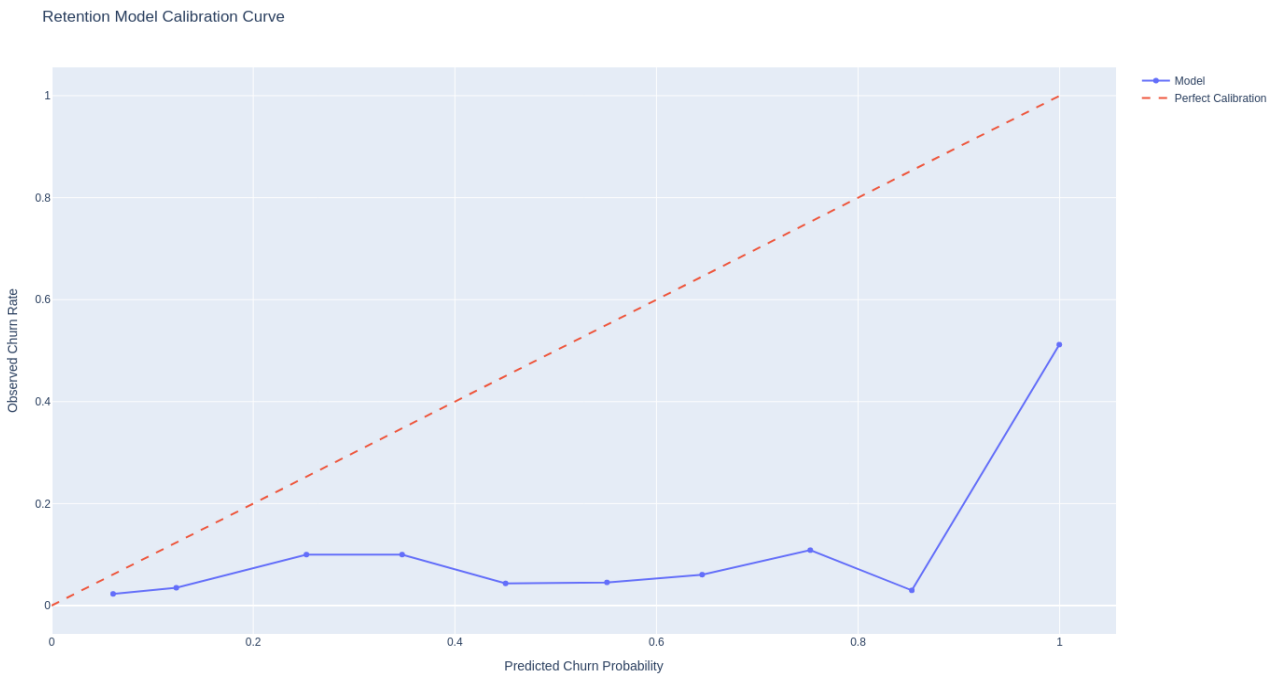


Acquisition Backtest Forecast

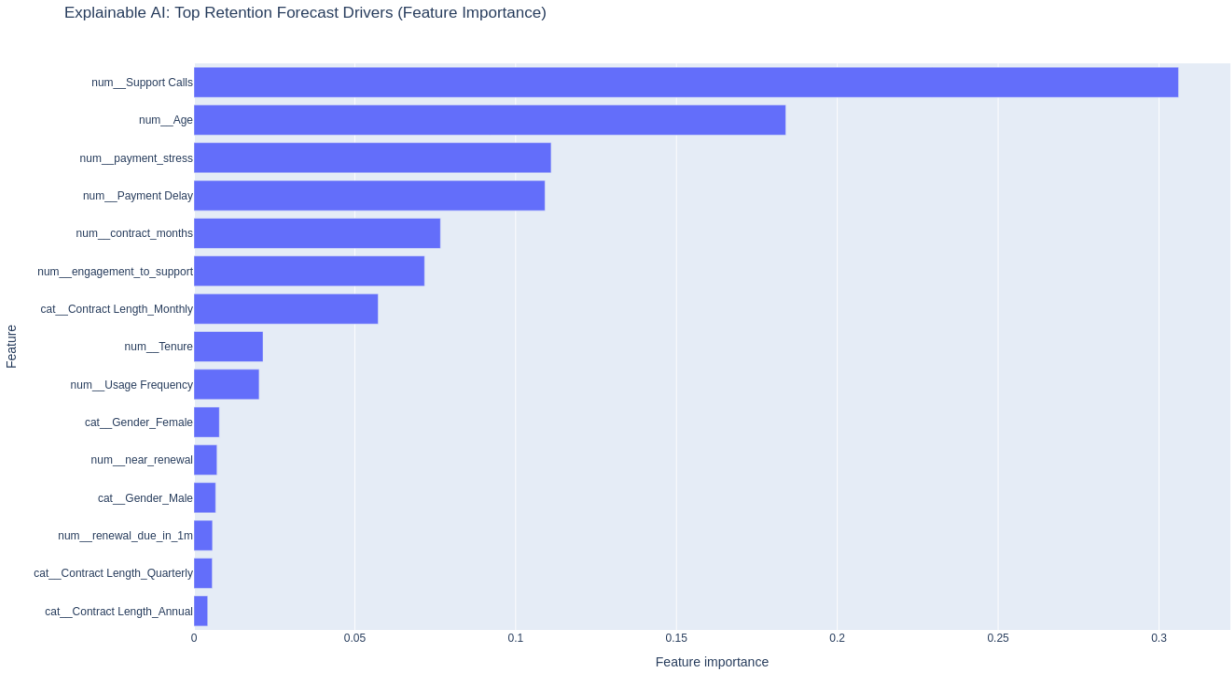
Retention Model Metrics



Retention Calibration

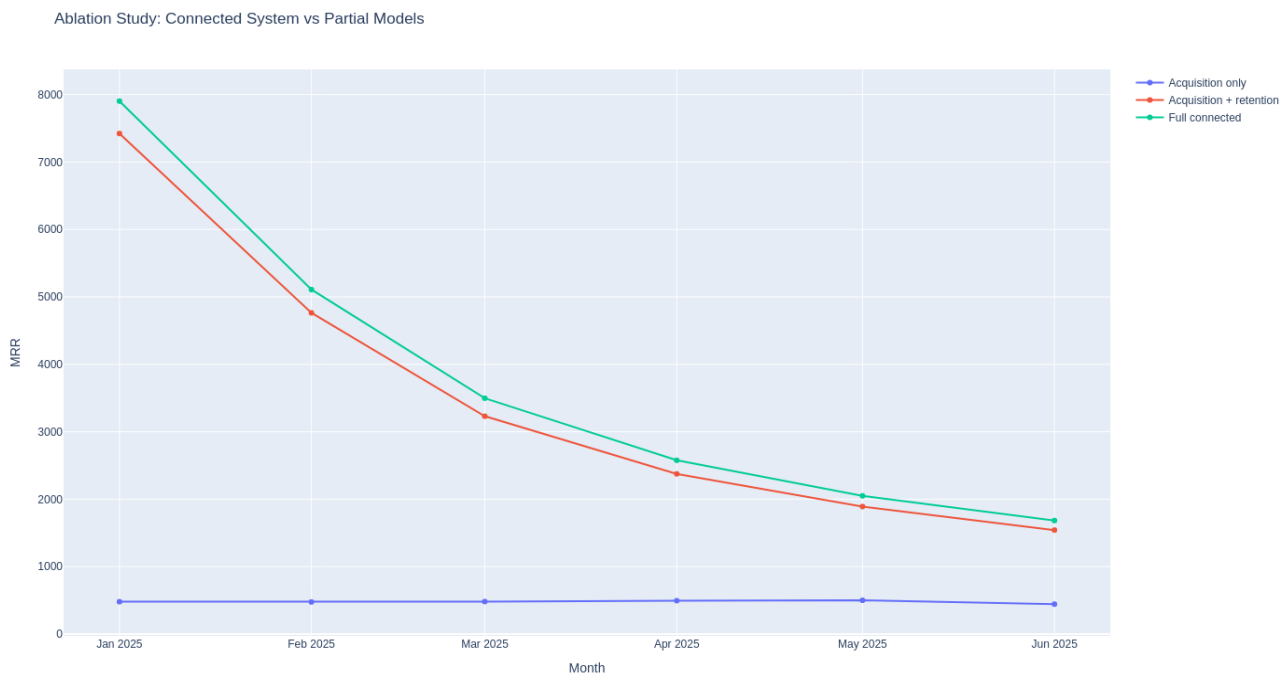


Retention Shap Importance

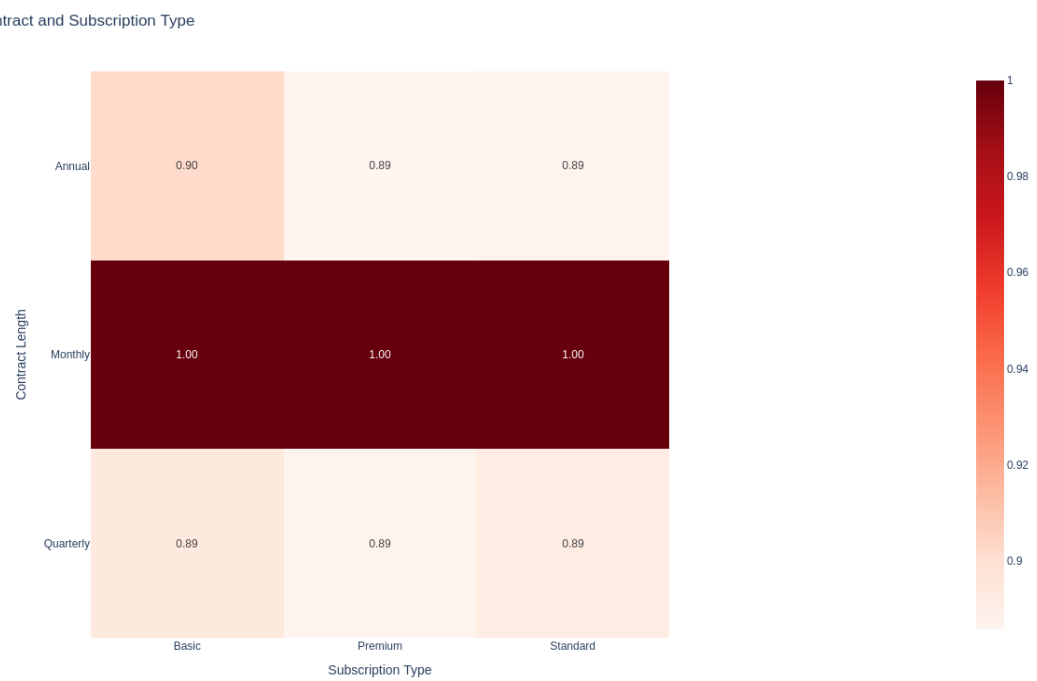


Connected Forecast Outputs

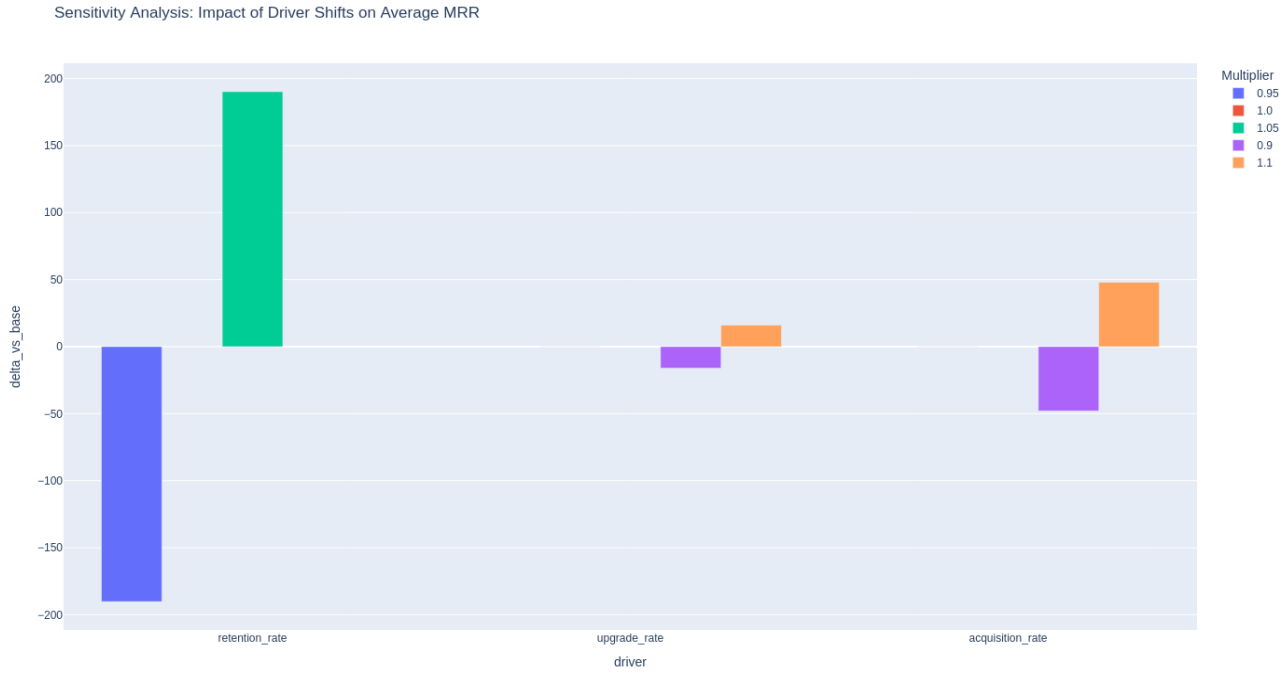
Connected Vs Isolated



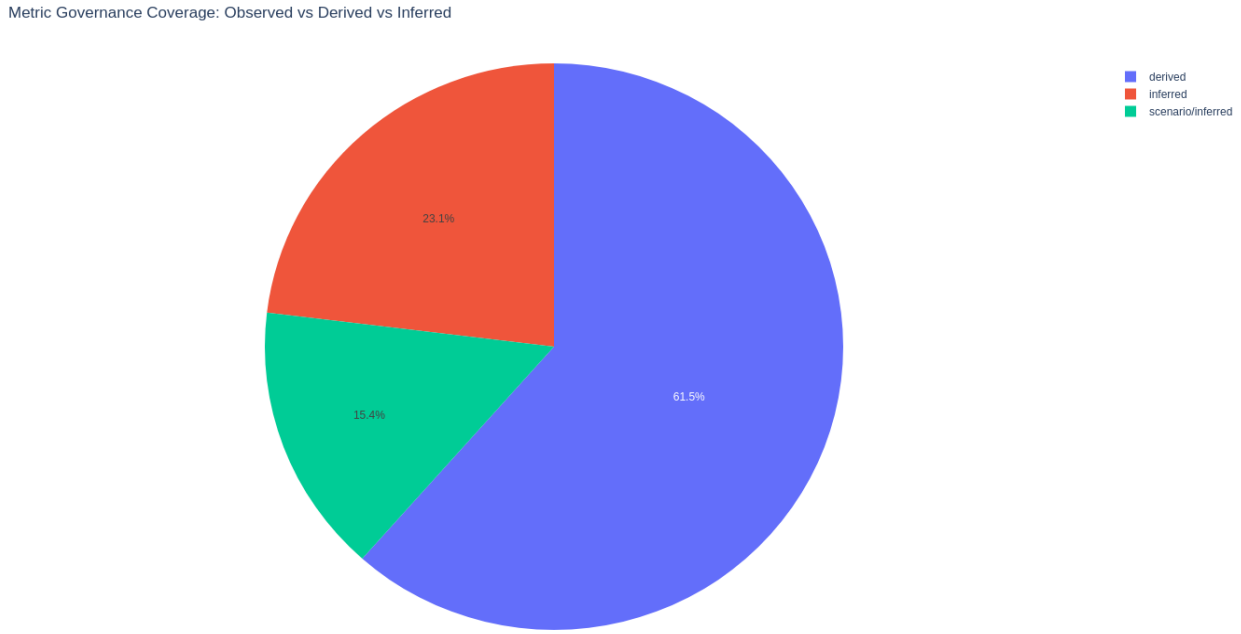
Cohort Churn Heatmap



Sensitivity Analysis



Metric Governance Coverage



8. Production AI tools architecture

- Metric governance and anomaly detection: Great Expectations / Soda / Monte Carlo plus semantic rule checks for drift in mapping, timing, and definitions.
- Feature store and orchestration: dbt + warehouse + scheduled training pipelines.
- Acquisition forecasting engine: rolling time-series model on subscription starts and pricing mix.
- Retention engine: calibrated churn model with cohort drill-down, threshold management, and probability calibration monitoring.
- Expansion engine: scenario layer now; upgrade to supervised ML only after event history is instrumented.
- Economic reconciliation service: deterministic ARR/MRR/ACV/TCV/CLV/CAC rollup layer with audit logs.
- Monitoring: forecast bias, model drift, feature drift, anomaly alerts, calibration shift, and feature-importance drift.
- Human governance loop: RevOps and Finance signoff on definitions and scenario assumptions.

9. File inventory

data/

- chart_data.json
- connected_forecast.csv
- connected_vs_isolated.csv
- contract_mapping_audit.csv
- metric_availability_matrix.csv
- retention_calibration.csv
- retention_cohort_summary.csv
- retention_feature_importance.csv
- retention_shap_importance.csv
- sensitivity_analysis.csv
- spend_proxy_outliers_sample.csv
- streaming_price_card.csv

docs/

- executive_summary.md
- methodology_notes.md

output/

- acquisition_backtest_forecast.png
- build_log.txt
- build_pid.txt
- cohort_churn_heatmap.png
- connected_forecast_outputs.png
- connected_vs_isolated.png
- data_inspection.txt
- driver_tree.png
- governance_audit_findings.json
- metric_governance_coverage.png

- model_summary.json
- pip_install.log
- pip_install_reportlab.log
- retention_calibration.png
- retention_metrics_full.json
- retention_metrics_strict.json
- retention_model_metrics.png
- retention_shap_importance.png
- sensitivity_analysis.png
- test_plot.png

code/

- build_revenue_driver_model.py
- generate_final_pdf.py
- inspect_data.py